

Solution Code - Module 3-Core Java - HashMapExample

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: Q25_HashMapExample.java

```
1 | import java.util.HashMap;
2 | import java.util.Scanner;
3 |
4 | public class Q25_HashMapExample {
5 |     public static void main(String[] args) {
6 |         HashMap<Integer, String> studentMap = new HashMap<>();
7 |         Scanner scanner = new Scanner(System.in);
8 |
9 |         System.out.println("=== Student ID to Name Mapping ===");
10 |        System.out.println("1. Add Student Entry");
11 |        System.out.println("2. Retrieve Student by ID");
12 |        System.out.println("3. Exit");
13 |
14 |        boolean running = true;
15 |        while (running) {
16 |            System.out.print("\nChoose an option (1, 2, 3): ");
17 |            if (scanner.hasNextInt()) {
18 |                int option = scanner.nextInt();
19 |                scanner.nextLine(); // consume newline
20 |
21 |                switch (option) {
22 |                    case 1:
23 |                        System.out.print("Enter Student ID (Integer): ");
24 |                        if (scanner.hasNextInt()) {
25 |                            int id = scanner.nextInt();
26 |                            scanner.nextLine(); // consume newline
27 |                            System.out.print("Enter Student Name: ");
28 |                            String name = scanner.nextLine().trim();
29 |
30 |                            if (!name.isEmpty()) {
31 |                                studentMap.put(id, name);
32 |                                System.out.println("Added: ID " + id + " -> " + name);
33 |                            } else {
34 |                                System.out.println("Name cannot be empty.");
35 |                            }
36 |                        } else {
37 |                            System.out.println("Invalid ID. ID must be an integer.");
38 |                            scanner.nextLine(); // clear invalid input
39 |                        }
40 |                        break;
41 |                    case 2:
42 |                        System.out.print("Enter Student ID to search: ");
43 |                        if (scanner.hasNextInt()) {
```

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```
44 |             int id = scanner.nextInt();
45 |             scanner.nextLine(); // consume newline
46 |             if (studentMap.containsKey(id)) {
47 |                 System.out.println("Student Found: ID " + id + " -> " +
studentMap.get(id));
48 |             } else {
49 |                 System.out.println("Student with ID " + id + " not found.");
50 |             }
51 |         } else {
52 |             System.out.println("Invalid ID. ID must be an integer.");
53 |             scanner.nextLine(); // clear invalid input
54 |         }
55 |         break;
56 |     case 3:
57 |         System.out.println("Exiting Student Map Manager.");
58 |         running = false;
59 |         break;
60 |     default:
61 |         System.out.println("Invalid option. Please choose 1, 2, or 3.");
62 |     }
63 | } else {
64 |     System.out.println("Invalid option. Please enter 1, 2, or 3.");
65 |     scanner.nextLine(); // clear invalid input
66 | }
67 | }
68 |
69 | scanner.close();
70 | }
71 | }
```